

VR Bus Training for Autistic Children: Building Independence in Public Transit

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ABSTRACT

This paper presents a VR application designed to help children and adolescents with autism spectrum disorder (ASD) practice public transportation skills. Users engage in tasks like finding the correct bus stop, boarding, checking in with a travel card, and exiting at the right stop. Built using Unity3D and Blender, the application offers realistic scenarios, interactive learning, and real-time feedback. By fostering independence, it addresses critical needs in K-12 education and life-skills training for students with ASD.

Index Terms: Autism, Virtual Reality, Education, Public Transportation, Training, Bus

1 INTRODUCTION

Independent navigation of public transportation is a critical skill that fosters autonomy, social inclusion, and participation in community life. However, children and adolescents with autism spectrum disorder (ASD) often face significant barriers to mastering these skills, including difficulties with spatial awareness, managing sensory input, and navigating social interactions during travel [2]. As a result, many municipalities allocate substantial resources to alternative transportation solutions, such as subsidized taxi services, which, while helpful, are not sustainable or empowering in the long term [5].

A recent systematic review of pedestrian navigation and public transit training interventions for youth with disabilities highlights the urgent need for innovative, realistic, and accessible training programs to improve these skills. The review identifies gaps in existing interventions, such as the lack of contextually relevant simulations, insufficient focus on youth with autism, and limited use of advanced technologies like virtual reality (VR). It also emphasizes the importance of developing tools that provide safe, engaging, and transferable learning environments to build independent navigation skills [3].

In response to these needs, we present a VR application designed to simulate the process of taking public transportation, specifically buses. This interactive system allows users to practice key tasks, including route planning with a realistic travel app, locating the correct bus stop, boarding the bus, choosing a seat, and disembarking at the right stop. By mirroring real-world public transportation systems, the application aims to address the gaps identified in prior research and support youth with autism in building the confidence and skills needed for independent travel. Our work contributes to the growing body of research exploring how VR technologies can enhance the lives of individuals with disabilities by meeting both practical and developmental needs.

Several studies have explored the use of VR for skill development in individuals with autism, focusing on different aspects of transportation and safety training. Simões et al. [6] presented an

proof-of-concept VR-based travel training intervention, emphasizing its potential for helping individuals with autism practice transportation tasks in a virtual environment. However, their system used abstract representations of buses and bus stops, lacked contextually realistic visual cues, and did not include other pedestrians or passengers, which are integral to real-world scenarios.

Miller et al. [4] designed a VR application for air travel training in children with autism, addressing anxieties associated with air travel but not focusing on the more frequent and accessible mode of public transportation, such as buses.

Our proposed VR application builds on these efforts by offering a comprehensive, task-specific simulation of bus travel tailored specifically to children and adolescents with ASD. We aim to enhance realism by incorporating country-specific visual cues, such as buses, bus stops, and transportation apps that closely resemble those used in the Danish public transportation system. Additionally, our system introduces pedestrians and other passengers into the virtual environment, adding a layer of social realism. By integrating features like interactive route planning, realistic boarding and ticketing, and contextually relevant feedback, our application addresses critical gaps in previous research. Furthermore, we combine developmental insights and pedagogical principles, such as scaffolding and real-time feedback, to foster independence and build transferable skills in a safe and engaging virtual environment.

2 DEMO DESCRIPTION

This demo presents a VR application developed to help children and adolescents with ASD practice essential public transportation skills in a safe and controlled environment. The application focuses on simulating the process of taking a bus, covering all key aspects of a typical journey, including route planning, locating the correct bus stop, boarding the bus, and exiting at the correct stop.

The VR application was developed using Unity3D, a widely used platform for creating interactive VR experiences. The 3D models used in the application, including the buses, bus stops, surrounding city environment, and interactive elements such as a mobile phone and travel card, were designed and created in Blender, an open-source 3D modeling and animation tool. These tools allowed for the creation of a realistic and visually engaging environment that mirrors real-world transportation systems.

The application is compatible with major VR headsets and controllers, providing an immersive and interactive experience for users. Navigation in the virtual environment is achieved through continuous locomotion using the analog stick on the VR controller. This method allows users to move naturally and explore the environment in a way that closely mirrors real-world movement.

2.1 The VR bus training application

The following steps outline the key features and functionalities of the application:

1. Tutorial and Objective Display:

- The experience begins at a large interactive billboard that provides a step-by-step tutorial on how to use the VR controls and complete tasks within the application (see figure 1).

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- Users learn to interact with the system using VR controllers. For example, pressing the left grip button brings up a virtual mobile phone that displays key information such as the bus number, direction of travel, and a clock showing the current time (see figure 1). Pressing the right grip button brings up a virtual travel card that is used to “purchase” a ticket on the bus (see figure 4).

2. Locating the Correct Bus Stop:

- The user spawns near two bus stops, both resembling real-world bus stops from the anonymous country (see figure 2). These stops include realistic signs and digital information screens.
- The bus stops display key information such as the bus number, direction of travel, and time remaining until the next bus arrives. Users must use these clues, similar to real-life navigation, to identify the correct bus stop based on the information provided on their virtual mobile phone.

3. Waiting and Boarding the Bus:

- Once at the correct bus stop, users wait for the bus to arrive. The buses display realistic signage indicating the bus number and destination (see figure 3).
- Upon arrival, users board the bus and are prompted to use the virtual travel card by pressing the right grip button to complete the ticket purchase. This step mirrors real-world interactions with transportation systems (see figure 4).

4. Choosing a Seat:

- Inside the bus, users can choose a seat by navigating to an available spot and interacting with it using the VR controller. Seats are designed to replicate real-world layouts, including options to sit next to other passengers or alone (see figure 5).

5. Monitoring Progress:

- The bus interior includes an information screen that displays the route progress and upcoming stops, similar to screens in actual buses.
- Users are encouraged to monitor the screen and observe the landmarks outside the window to identify their destination stop.

6. Exiting the Bus:

- When the destination stop is approaching, users must press the bus’s stop button to signal the driver.
- Upon arriving at the stop, users exit the bus and must remember to check out with their virtual travel card using the right grip button. A congratulatory message then appears on the virtual phone to indicate successful completion of the task.



Figure 1: The tutorial billboard and the mobile travel app



Figure 2: The bus stop with a departure information screen resembling its real-world counterpart



Figure 3: The bus resembling a real-world bus from an anonymous country, complete with information screens, and the mobile app displaying the next sub-objective



Figure 4: The student travel card and the ticket machine, replicating real-world ticketing machines found on buses in the anonymous country.



Figure 5: Inside the bus. Player can pick a place to sit by pressing the button next to the sit they want to pick

3 CONCLUSION

This paper presents a VR application designed to help autistic children and adolescents develop essential public transportation skills in a realistic, immersive, and safe environment. By simulating the entire journey of taking a bus, the application addresses significant challenges faced by individuals with ASD, including navigating unfamiliar environments, managing anxiety, and mastering sequential tasks required for independent travel. Through interactive features, pedagogical design principles, and potential for real-world application, this VR solution represents a meaningful step toward fostering independence and inclusion for students with ASD.

The application emphasizes interactive learning by integrating real-time feedback and guidance throughout the experience. For instance, if users attempt to board the wrong bus, the system provides a notification and prompts them to review the travel information on the virtual mobile phone. Similarly, if users forget to check in or out with their virtual travel card, the application issues reminders, reinforcing essential behaviors. Tasks are divided into smaller, manageable objectives that are displayed on the mobile phone, allowing users to track their progress and stay focused on their goals. This approach enables learners to engage in trial-and-error practices in a low-stakes environment, promoting confidence and motivation.

Aligned with pedagogical principles of scaffolding and embodied learning [1], the VR application provides users with structured opportunities to practice critical life skills through repetition. Positive reinforcement mechanisms, such as the congratulatory message displayed upon task completion, enhance motivation and engagement. By offering a safe and controlled environment, the application minimizes the anxiety often associated with learning new skills, making it a valuable resource for educators and therapists

working with individuals with ASD. This immersive and interactive approach supports not only the development of transportation skills but also broader competencies related to autonomy and daily living.

The application's real-world relevance is a major strength, addressing the reliance on costly taxi services that many students with ASD require due to their challenges in navigating public transportation. By simulating real-world bus travel, the application equips users with practical skills that can be directly transferred to real-world scenarios. The contextual realism, including features that replicate actual transportation systems such as bus stops, signage, and boarding procedures, ensures that users gain familiarity and confidence in their local environment. This approach enhances the transferability of skills while reducing costs for municipalities and schools and fostering greater community participation for individuals with ASD.

To maximize its impact, the application can be implemented in K-12 educational programs to support life-skills training for students with ASD. Schools can integrate the VR system into special education curricula or individualized education plans, providing regular opportunities for students to practice transportation skills in a virtual setting. Educators and therapists can monitor students' progress and provide tailored support, gradually introducing real-world practice as confidence builds in the virtual environment. For example, weekly training sessions could be combined with group activities or role-playing exercises to enhance social skills alongside transportation training. Collaborations with local transportation authorities could also enable field trips that allow students to apply their virtual training to real-life scenarios under supervised conditions.

The effectiveness of the VR application can be evaluated through a combination of qualitative and quantitative methods. Metrics such as task completion rates, accuracy of task execution, and reduction in errors can provide measurable insights into skill acquisition. Additionally, pre- and post-intervention assessments can evaluate the impact of the system on students' confidence, anxiety levels, and readiness to transition from virtual practice to real-world application. Longitudinal studies that track students' progress over time, particularly as they apply their skills in real-world contexts, could offer deeper insights into the long-term benefits of the tool.

Looking forward, this VR application could be expanded to include additional forms of public transportation, such as trains and subways, broadening its scope and applicability. Customizable features, such as difficulty levels or adaptive guidance tailored to individual users' needs, could further enhance its accessibility and effectiveness. The continued development and implementation of such tools will rely on collaboration between educators, researchers, and technology developers to ensure that the unique needs of students with ASD are met comprehensively.

This VR application demonstrates the transformative potential of immersive technologies in addressing critical challenges faced by individuals with ASD. By providing a realistic, engaging, and structured environment to practice public transportation skills, the application empowers students to become more independent and self-reliant in their daily lives. Its integration into K-12 educational settings represents a significant step toward fostering autonomy and inclusion for students with ASD, while also offering practical benefits to schools, families, and communities.

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